

## COTTONWOOD PASS, CO, USA

WMO: 726391

Lat: 38.782N

Lon: 106.218W

Elev: 9827

StdP: 10.17

Time Zone: -7.00 (NAM)

Period: 05-19

WBAN: 00385

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD<br>to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|-------|-----|------|-----------------------|------|------|------|--------------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%   |      |                          |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                     | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                      | (o)  | (p)   |
| 12            | 2.8        | 8.6 | -36.5                         | 1.0 | 38.4 | -27.3 | 1.7 | 36.8 | 39.8                  | 31.5 | 34.8 | 29.8 | 5.6                      | 270  | 0.627 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 21.3                   | 80.6            | 47.6 | 77.2 | 47.2 | 73.4 | 46.6 | 54.4                | 65.6 | 53.1 | 65.0 | 51.8 | 64.3 | 5.5                  | 340  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 51.6                            | 82.6 | 58.6 | 48.5 | 73.2 | 57.4 | 46.1 | 66.8 | 57.4 | 27.2          | 65.6 | 26.1 | 64.6 | 25.3 | 64.4 | 60.3              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |       |            |       |            |       |      |     |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 31.1          | 25.8 | 21.5 | DB                         | -1.0 | 85.8               | 2.8 | 1.7                                                | -3.0 | 87.0       | -4.7  | 88.0       | -6.3  | 88.9       | -8.4  | 90.1 | (4) |
| (5)               |      |      | WB                         | -3.7 | 57.9               | 5.3 | 1.8                                                | -7.5 | 59.2       | -10.6 | 60.2       | -13.6 | 61.2       | -17.5 | 62.5 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 44.5        | 27.8  | 27.8  | 34.7  | 39.6  | 48.1  | 61.2  | 65.0  | 62.5  | 56.6  | 45.0  | 36.7  | 27.5  | (6)  |  |
|                                                                        | DBStd       | 15.68       | 8.80  | 9.08  | 8.67  | 7.71  | 8.16  | 7.00  | 4.15  | 4.20  | 6.48  | 8.72  | 9.30  | 9.50  | (7)  |  |
|                                                                        | HDD50       | 3570        | 689   | 622   | 477   | 318   | 136   | 10    | 0     | 0     | 21    | 197   | 401   | 699   | (8)  |  |
|                                                                        | HDD65       | 7610        | 1154  | 1042  | 941   | 761   | 525   | 146   | 51    | 98    | 257   | 621   | 850   | 1164  | (9)  |  |
|                                                                        | CDD50       | 1544        | 0     | 0     | 1     | 6     | 76    | 346   | 466   | 387   | 219   | 41    | 2     | 0     | (10) |  |
|                                                                        | CDD65       | 110         | 0     | 0     | 0     | 0     | 0     | 33    | 52    | 20    | 5     | 0     | 0     | 0     | (11) |  |
|                                                                        | CDH74       | 665         | 0     | 0     | 0     | 0     | 0     | 231   | 304   | 101   | 29    | 0     | 0     | 0     | (12) |  |
|                                                                        | CDH80       | 65          | 0     | 0     | 0     | 0     | 0     | 27    | 34    | 3     | 1     | 0     | 0     | 0     | (13) |  |
| (14) Wind                                                              | WSAvg       | 7.8         | 8.7   | 10.3  | 9.0   | 8.6   | 7.0   | 6.7   | 5.3   | 5.4   | 6.2   | 8.0   | 8.9   | 9.1   | (14) |  |
| (15) Precipitation                                                     | PrecAvg     | 11.3        | 0.4   | 0.5   | 0.8   | 1.2   | 1.1   | 0.8   | 1.7   | 2.1   | 1.1   | 0.9   | 0.6   | 0.4   | (15) |  |
|                                                                        | PrecMax     | 14.0        | 0.9   | 1.2   | 1.9   | 2.7   | 3.1   | 1.5   | 3.4   | 3.9   | 2.0   | 2.3   | 1.3   | 2.3   | (16) |  |
|                                                                        | PrecMin     | 7.6         | 0.1   | 0.1   | 0.2   | 0.2   | 0.1   | 0.3   | 0.3   | 0.7   | 0.4   | 0.2   | 0.2   | 0.2   | (17) |  |
|                                                                        | PrecStd     | 1.9         | 0.2   | 0.4   | 0.4   | 0.8   | 0.8   | 0.4   | 0.9   | 0.7   | 0.5   | 0.5   | 0.3   | 0.5   | (18) |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 50.2  | 50.5  | 59.4  | 63.3  | 72.4  | 82.4  | 82.5  | 80.6  | 78.6  | 67.9  | 59.0  | 49.8  | (19) |  |
|                                                                        |             | MCWB        | 33.0  | 32.3  | 36.2  | 39.1  | 42.8  | 46.4  | 48.7  | 47.9  | 46.0  | 41.8  | 36.7  | 31.2  | (20) |  |
|                                                                        | 2%          | DB          | 45.2  | 45.8  | 54.2  | 59.0  | 68.1  | 79.5  | 80.8  | 77.0  | 73.1  | 63.4  | 54.4  | 45.2  | (21) |  |
|                                                                        |             | MCWB        | 30.3  | 30.7  | 34.4  | 36.8  | 41.5  | 46.0  | 48.7  | 47.5  | 44.8  | 40.2  | 35.5  | 29.4  | (22) |  |
|                                                                        | 5%          | DB          | 42.5  | 43.1  | 50.3  | 54.7  | 64.0  | 76.8  | 77.4  | 73.3  | 70.4  | 60.8  | 51.9  | 42.6  | (23) |  |
|                                                                        |             | MCWB        | 29.0  | 29.1  | 32.7  | 35.5  | 40.3  | 45.4  | 48.7  | 47.5  | 44.5  | 39.4  | 34.4  | 28.7  | (24) |  |
|                                                                        | 10%         | DB          | 37.4  | 39.2  | 46.1  | 52.2  | 61.3  | 72.9  | 73.5  | 72.0  | 67.9  | 56.6  | 48.1  | 38.9  | (25) |  |
|                                                                        |             | MCWB        | 26.8  | 27.6  | 31.1  | 34.6  | 39.6  | 44.3  | 48.3  | 47.6  | 44.2  | 38.4  | 32.7  | 27.5  | (26) |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 35.2  | 34.9  | 38.0  | 41.6  | 46.4  | 52.9  | 57.8  | 55.9  | 52.7  | 47.8  | 39.8  | 34.4  | (27) |  |
|                                                                        |             | MCDB        | 44.2  | 44.9  | 53.7  | 54.4  | 59.6  | 65.9  | 68.8  | 66.0  | 63.3  | 56.3  | 50.9  | 41.9  | (28) |  |
|                                                                        | 2%          | WB          | 32.4  | 32.3  | 35.4  | 39.5  | 44.4  | 50.6  | 55.2  | 54.0  | 50.5  | 44.5  | 37.1  | 32.2  | (29) |  |
|                                                                        |             | MCDB        | 41.2  | 41.7  | 49.1  | 52.8  | 59.3  | 66.2  | 66.1  | 64.9  | 61.2  | 55.3  | 50.0  | 39.9  | (30) |  |
|                                                                        | 5%          | WB          | 30.5  | 30.8  | 34.0  | 37.4  | 43.0  | 49.1  | 53.8  | 52.6  | 48.8  | 42.3  | 35.4  | 30.4  | (31) |  |
|                                                                        |             | MCDB        | 39.4  | 39.9  | 47.0  | 50.7  | 58.3  | 65.7  | 65.7  | 64.0  | 61.2  | 53.8  | 47.7  | 39.1  | (32) |  |
|                                                                        | 10%         | WB          | 28.4  | 29.2  | 32.4  | 35.7  | 41.5  | 47.7  | 52.5  | 51.3  | 47.3  | 40.4  | 33.7  | 28.8  | (33) |  |
|                                                                        |             | MCDB        | 36.8  | 38.2  | 44.4  | 48.0  | 55.6  | 65.8  | 65.2  | 63.2  | 60.1  | 52.9  | 46.5  | 37.6  | (34) |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 12.8  | 13.0  | 15.4  | 17.3  | 18.8  | 21.5  | 21.3  | 19.4  | 19.2  | 16.6  | 14.3  | 11.9  | (35) |  |
|                                                                        |             | MCDBR       | 13.8  | 14.4  | 18.0  | 20.1  | 21.3  | 23.9  | 23.5  | 21.3  | 21.3  | 19.0  | 16.1  | 12.7  | (36) |  |
|                                                                        | 5% WB       | MCWBR       | 8.0   | 7.8   | 8.2   | 8.9   | 8.4   | 8.6   | 7.4   | 7.3   | 7.5   | 8.2   | 7.5   | 7.0   | (37) |  |
|                                                                        |             | MCWBR       | 12.4  | 12.6  | 16.4  | 18.9  | 20.5  | 22.8  | 20.5  | 18.8  | 18.0  | 17.1  | 15.0  | 11.5  | (38) |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        | taub        | 0.206 | 0.214 | 0.237 | 0.258 | 0.267 | 0.270 | 0.307 | 0.294 | 0.256 | 0.239 | 0.217 | 0.208 | (40) |  |
|                                                                        |             | taud        | 2.497 | 2.461 | 2.461 | 2.514 | 2.597 | 2.618 | 2.537 | 2.559 | 2.703 | 2.685 | 2.635 | 2.530 | (41) |  |
|                                                                        | Ebn at Noon | Ebn at Noon | 322   | 330   | 329   | 325   | 321   | 318   | 305   | 309   | 319   | 318   | 316   | 314   | (42) |  |
|                                                                        |             | Edh at Noon | 23    | 28    | 32    | 32    | 30    | 30    | 32    | 31    | 25    | 23    | 21    | 21    | (43) |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      | RadAvg      | 875   | 1178  | 1609  | 1882  | 2040  | 2314  | 2031  | 1832  | 1668  | 1277  | 932   | 751   | (44) |  |
|                                                                        | RadStd      | RadStd      | 44    | 66    | 120   | 117   | 155   | 139   | 111   | 88    | 103   | 90    | 61    | 49    | (45) |  |

## Historical Trends

|                              | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (29 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | +131  | +36   |

Nomenclature: See separate page